

Sound-Operated DEVICE-CONTROL SYSTEM

ARJAV BAVARVA, RAHUL VYAS, PARTH GANGADIA AND FAIZAL PANAVADHU

In movies like *Ironman* and *Players*, it is shown that appliances are operated by human speech

or sound. Similar concept is implemented in this project. The system responds to pre-stored sound and performs particular functions. That is, command signals are first stored in the database and when you run the project, a real-time sound or speech signal is compared with the signals stored in the database. If signals match, an appliance/

load is turned on/off. For instance, sound command 'Switch on the fan' is used to operate the fan.

This project has vital applications especially for handicaps. It can also be used to provide security and authentication.

Block diagram

The project can be developed by various methods using different algorithms like biometrics, fuzzy logic, neural networks and hidden Markov. The easiest and most efficient method is cross-correlation

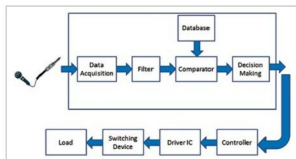


Fig. 1: Block diagram of the project with connections between different modules

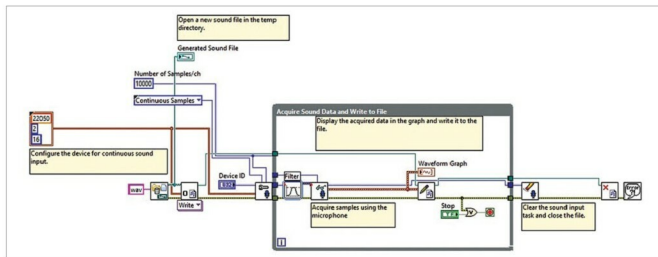


Fig. 2: Wave file creation VI

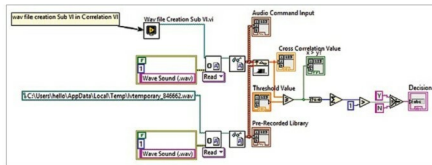


Fig. 3: Comparison of wave file

algorithm implemented by LabVIEW software. Fig. 1 shows the block diagram of the project with connections between different modules. Various processes like speech acquisition, speech filtering, speech comparison and decision making are easy to program in LabVIEW because of its user-friendly graphical programming environment.

A predefined command is stored